

Capstone Project: Data-Driven Surveillance:
Identifying Early Chronic Disease Trends in the US Through Medicare Part D Prescriptions
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Abstract

This capstone project explores geographic disparities in opioid prescribing among Medicare Part D beneficiaries aged 65 and older. It aims to identify counties with unusually high or low prescribing rates and examine associations with demographic factors such as income, rurality, and race.

Using a retrospective, observational design, aggregated opioid prescribing data (2013–2023) from the Centers for Medicare & Medicaid Services were analyzed. Key metrics included total opioid claims, beneficiary counts, and cost-sharing rates. Data were normalized to prescribing rates per 1,000 beneficiaries and visualized using R to identify trends. Counties with suppressed or incomplete data were marked as gaps.

Findings show significant regional variation in opioid prescribing, with higher rates in rural and low-income counties. Predominantly white counties tended to have greater prescription volumes. Some counties deviated from expected patterns, suggesting the influence of local policy, provider behavior, or unmeasured factors. Missing data—especially in rural areas—limited complete analysis.

County-level variation reveals the need for tailored public health interventions. Data visualization effectively identified trends and blind spots. Improving data completeness and focusing on high-risk areas are key for equitable policy development.

Introduction

The opioid crisis is often portrayed in broad strokes—images of overdose deaths, grim statistics on addiction, and national headlines about lawsuits against pharmaceutical giants. However, beneath the surface lies a complex and intensely geographic story. Where a person's life can significantly shape their exposure to opioids, their chances of being prescribed these drugs, and ultimately, their risk for dependency or overdose. My capstone project focuses on this critical and underexplored dimension: geographic disparities in opioid prescribing among older adults enrolled in Medicare Part D.

This topic matters not just because opioids remain a persistent public health crisis, but because older adults, particularly those aged 65 and above, have unique vulnerabilities. They often suffer from multiple chronic pain conditions, rely heavily on prescriptions, and may be overlooked in conversations that are typically centered on younger populations. By examining where opioid prescriptions are most concentrated and who is receiving them, we gain key insights into structural disparities in healthcare, prescriber behavior, and the social determinants of health.

Understanding geographic variation in prescribing is not just an academic exercise; it is essential for developing targeted interventions. Policymakers and public health professionals cannot address opioid misuse effectively without knowing where the problems are most acute. The stakes are high when too little access to pain management can result in unnecessary suffering, while too much access can lead to addiction, hospitalization, or death.

This project stands at the intersection of epidemiology, health policy, and informatics. Using publicly available Medicare Part D prescription data, my goal is to identify patterns in opioid prescribing among adults aged 65 and older across different U.S. counties. By visualizing these trends and pairing them with demographic data, we can gain a deeper understanding of which communities are most affected and why.

Background: The Scope of the Opioid Epidemic

Since the late 1990s, the United States has been grappling with a devastating opioid epidemic. Driven first by the over-prescription of pain relievers and later by the influx of heroin and synthetic opioids like fentanyl, this public health emergency has taken a staggering toll. According to the Centers for Disease Control and Prevention (CDC), nearly 645,000 people died from opioid overdoses between 1999 and 2021 (CDC, 2022). While much of the early focus was on younger and middle-aged populations, a growing body of research shows that older adults are increasingly affected.

Older adults often live with chronic conditions that require pain management, arthritis, osteoporosis, and post-surgical recovery, and may be prescribed opioids at higher rates than younger patients (Schieber et al., 2019). Additionally, age-related changes in drug metabolism, polypharmacy, and cognitive decline put older adults at higher risk of complications from opioids (Hudson et al., 2017; Gomes et al., 2011). Yet, despite these risks, they are frequently underrepresented in both research and intervention strategies aimed at reducing opioid misuse.

Medicare Part D and the Importance of Prescribing Data

Medicare Part D, introduced in 2006, provides prescription drug coverage to millions of Americans, primarily those over 65 (CMS, 2024). It provides a rich dataset that can be used to

analyze patterns in drug prescribing across demographics and geographic regions (Centers for Medicare & Medicaid Services, 2024). The Centers for Medicare & Medicaid Services (CMS) makes anonymized, aggregated data available to researchers and the public, allowing for in-depth investigation into prescribing trends.

Previous studies have used Medicare Part D data to explore provider-level prescribing habits and temporal trends in opioid use (Spivack & Wu, 2024). However, fewer have focused explicitly on the geographic disparities at the county level, especially among older adults. Understanding how prescribing varies by location can help identify communities where prescribing may be excessive or insufficient, informing both clinical and policy decisions (Chan & Miller, 2022; Dasgupta et al., 2018).

Gaps in the Current Literature

Although the opioid epidemic has been extensively studied, several vital gaps remain, particularly for older adults. Much of the existing research has concentrated on younger populations or national trends, leaving behind more localized and demographically nuanced insights.

Li, Chen, and Patel (2025) have highlighted broad geographic and socioeconomic disparities in opioid prescribing within the Medicare Part D population, showing regional variation across states. However, their analysis does not disaggregate these disparities down to the county level, limiting the specificity of intervention strategies. Similarly, Spivack and Wu (2024) analyzed prescribing trends over nearly a decade and documented persistent regional differences; however, they did not integrate social determinants such as race, income, or education.

Chan and Miller (2022) advanced the discussion by examining opioid use disorder (OUD) rates among older adults at the county level. Their spatial modeling approach revealed important insights into geographic inequities, yet their focus was on OUD outcomes rather than prescribing behavior itself. Meanwhile, Essien et al. (2021) underscored significant racial and ethnic disparities in opioid prescribing among Medicare beneficiaries, with Black and Hispanic older adults less likely to receive prescriptions than white patients. However, their study did not explore how these racial disparities intersect with geography.

Additional gaps emerge in research on clinical practices and systems-level drivers. Freeman and Zhao (2023) explored how continuity in opioid prescribing correlates with better outcomes for older adults but did not investigate where such continuity is most (or least) common geographically. Moyo and Harris (2024) examined the complexities of prescribing and deprescribing opioids in older patients, yet their findings were not tied to location-based trends.

Structural barriers such as formulary exclusions have also been shown to affect opioid prescribing. Alexander and Saloner (2020) found that limited access to non-opioid alternatives may unintentionally increase opioid prescriptions in Medicare, but their work stopped short of analyzing regional disparities. Likewise, policy-based studies such as those by Meara et al. (2016) and Bao et al. (2019) reviewed the effectiveness of state programs and prescription drug monitoring systems. However, they did not address how county-level characteristics might modulate the impact of these programs.

In summary, these studies offer crucial insights into opioid prescribing, racial and socioeconomic disparities, and policy effects. However, few connect all three dimensions at the county level, particularly for the aging population. This project addresses those gaps by:

- Focusing specifically on Medicare beneficiaries aged 65 and older.

- Analyzing opioid prescribing at the county level.
- Integrating demographic and structural variables such as race, rurality, income, and education.

Based on the literature, several patterns are expected to emerge:

- Higher prescribing rates in rural counties are often due to limited access to non-opioid pain treatment and a higher prevalence of manual labor injuries (Li et al., 2025; Dasgupta et al., 2018).
- Elevated prescribing in low-income or low-education counties reflects healthcare disparities (Chan & Miller, 2022; Essien et al., 2021).
- Racial disparities in prescribing, with majority-white counties receiving more opioid prescriptions than majority-Black or Hispanic ones, as indicated by previous research.

However, one of the strengths of this study lies in its potential to identify anomalies—counties that defy expected trends—offering opportunities for deeper inquiry into what policies, cultural factors, or clinical behaviors drive those exceptions.

Research Problem

The central research problem addressed in this study is the under-exploration of county-level disparities in opioid prescribing among older adults.

In recent years, there has been growing recognition of the role that geographic context plays in shaping health outcomes. The concept of "place-based health" suggests that where people live influences their health as much as, if not more than, their individual choices or genetics (Beyer et al., 2011; Diez Roux, 2016; Dasgupta, Beletsky, & Ciccarone, 2018). This idea is particularly relevant in the context of opioid prescribing, where factors like provider

density, state policies, rural versus urban status, and socioeconomic conditions intersect in complex ways.

Despite the importance of geography, older adults have often been overlooked in discussions of this topic. There is a pressing need to examine how opioid prescribing practices vary across counties for Medicare Part D beneficiaries aged 65 and older. Are certain counties prescribing opioids at significantly higher rates? Are there correlations between these rates and community characteristics, such as income, rurality, or race? Are these disparities reflective of fundamental differences in need, or do they point to systemic issues in prescribing practices?

These are not just academic questions—they have real-world implications. High prescribing rates may indicate overuse or insufficient use of alternative pain management strategies. Conversely, low prescribing rates may indicate barriers to accessing appropriate care. Either scenario can lead to poor health outcomes and decreased quality of life for older adults.

My project aims to identify and visualize these disparities, contributing to the broader understanding of how place and policy influence opioid use in vulnerable populations.

Objectives and Research Question

This capstone project has four primary objectives:

1. Quantify county-level variation in opioid prescribing rates among Medicare Part D beneficiaries aged 65 and older across the United States.
2. Identify counties with prescribing rates significantly above or below the national average.
3. Explore associations between opioid prescribing rates and county-level demographic factors such as income, rurality, education, and race/ethnicity.

4. Visualize the findings using data dashboards and maps to make the data accessible to both academic and public health audiences.

Research Question

The central research question guiding this project is:

"How do opioid prescribing rates for Medicare Part D beneficiaries aged 65 and older vary across U.S. counties, and what demographic and structural factors are associated with these geographic disparities?"

This question is intentionally broad to allow for the discovery of both expected and unexpected patterns in the data. It allows us to investigate not only the where of opioid prescribing, but also the why—an essential step toward developing evidence-based interventions.

Methodology

This retrospective, observational study examined geographic disparities in opioid prescribing among Medicare Part D beneficiaries aged 65 and older across the United States. The research aimed to identify high-prescribing regions, visualize geographic variation in prescribing intensity, assess trends over time, and pinpoint areas where missing or suppressed data may signal blind spots in surveillance. Using de-identified data from the Centers for Medicare & Medicaid Services (CMS), this analysis employed advanced statistical, mapping, and data-cleaning techniques within the R programming environment to develop county- and state-level opioid prescribing metrics. The principles of transparency, reproducibility, and public health relevance guided the development of the methodology.

Population

The target population consisted of Medicare Part D beneficiaries aged 65 years and older who received at least one opioid prescription between 2013 and 2023. This group was selected due to their elevated risk of chronic pain, multimorbidity, hospitalizations, and overall adverse drug events compared to younger populations. Older adults are more likely to receive long-term opioid therapy for chronic musculoskeletal pain, arthritis, or post-surgical recovery (Gomes et al., 2011). However, they are also more vulnerable to complications such as falls, sedation, respiratory depression, and hospitalizations resulting from opioid use (Hudson et al., 2017).

Although provider prescribing behavior was indirectly assessed through aggregate claims data, the beneficiary served as the unit of analysis. Focusing on this age group enabled the research to examine opioid prescribing in a population where opioid-related harms are often underreported or overlooked in favor of younger, high-profile overdose populations.

Study Site

The geographic scope of the study included all 50 U.S. states, the District of Columbia, Puerto Rico, and the U.S. Virgin Islands. Geographic analysis was conducted at both the county and state levels, allowing researchers to examine regional differences in prescribing practices. The rationale for dual-level geographic analysis stems from the need to provide both a broad national overview and a granular, community-focused understanding of opioid prescribing patterns (Dasgupta et al., 2018).

County-level analysis was prioritized when possible because public health interventions, such as naloxone distribution, prescription monitoring programs, and general population health needs, are often implemented at the local level (Dasgupta et al., 2018). Geographic variability in

health care delivery systems and the social determinants of health made the county an appropriate level for public health inquiry.

Sample and Eligibility Criteria

The dataset was sourced from the publicly available Centers for Medicare & Medicaid Services (CMS) Medicare Part D Prescriber Public Use Files (CMS, 2024). These files include de-identified, aggregated metrics on opioid prescriptions at the county level, including total claims, total beneficiaries, and total drug costs. Data from 2013 to 2023 were used to examine trends and cumulative burden.

Inclusion criteria required records to (a) contain a valid county or state geographic identifier, (b) reflect a beneficiary population aged 65 and older, and (c) report at least one opioid prescription claim during the specified timeframe. Exclusion criteria included (a) suppressed or missing data with fewer than 11 claims, and (b) records not associated with U.S. states or territories. Counties with unreported data were acknowledged as gaps in the dataset and were visually flagged on spatial maps.

Variables and Measures

Primary variables extracted from the CMS files included:

- GE65_Tot_Clms: Total number of opioid claims
- GE65_Tot_Benes: Total number of unique beneficiaries aged 65+
- GE65_Tot_Drug_Cst: Total drug cost associated with the claims
- LIS_Bene_Cst_Shr: Cost share for Low-Income Subsidy (LIS) beneficiaries
- NonLIS_Bene_Cst_Shr: Cost share for non-LIS beneficiaries

Derived variables were calculated to support comparative analysis:

- $\text{Claims per 1,000 beneficiaries} = (\text{Total Claims} / \text{Total Beneficiaries}) * 1,000$
- $\text{Prescribing Rate} = (\text{Opioid Claims} / \text{Total Claims}) * 100$

These variables allowed for normalization across counties and identification of intensity and cost differences. These metrics were calculated at both the county and state levels using R's *dplyr* package. Long-acting opioid prescribing and cost share differences between LIS and non-LIS populations were also explored, where data permitted.

Quality Control

To ensure data accuracy and reliability, multiple quality control measures were implemented throughout the analysis. Missing values were identified and excluded using the *na.rm = TRUE* argument in R to prevent distortion in aggregate calculations. Manual spot-checking was conducted to evaluate outliers and extreme values, particularly at the county level. All data transformations and aggregations were carefully documented and version-controlled using R scripts to maintain consistency and reproducibility. Spatial join techniques and visual inspections were used to verify proper alignment between prescribing data and geographic shapefiles.

Additionally, datasets were cross-validated using official CMS documentation and summaries from previous years, where available. Counties with suppressed or unavailable data were flagged and annotated on choropleth maps to highlight areas of uncertainty in spatial analyses. All scripts and analyses were documented and version-controlled to ensure reproducibility.

Data Transformation and Statistical Analysis

Data management and analysis were performed using R (version 2024.04.2+764). The primary packages included *dplyr* (Wickham et al., 2019) for data wrangling, *ggplot2* (Wickham, 2016) for data visualization, *sf* (Pebesma, 2018) for spatial data geometry, and *tmap* for thematic mapping. Descriptive statistics and trend visualizations were used to assess temporal shifts in prescribing rates (2013–2023). Line charts tracked the national average prescribing rate by year. Bar charts ranked states and counties by prescribing intensity.

County-level trends over time were plotted to assess variance and identify emerging hotspots. The top 10 counties with the highest mean opioid prescribing rates were identified and compared. Where multiple records existed per county per year, data were averaged to produce clean trendlines. When totals were available but individual components (e.g., LIS cost) were missing, these rows were excluded from cost comparisons but retained for rate calculations.

To normalize for differing beneficiary counts, rates were expressed per 1,000 beneficiaries or as percentages of total claims, rather than using raw counts. This was crucial for ensuring interpretability and fairness in cross-county and cross-state comparisons. Future analyses may incorporate inferential statistics, such as ANOVA or multivariable regression, to evaluate associations between opioid prescribing rates and demographic or socioeconomic indicators.

Visualization and Geospatial Analysis

The study relied heavily on visual analysis to communicate geographic and temporal patterns. The following tools and packages were used:

- *ggplot2*: For trend lines, bar charts, and ranking visualizations. Notable outputs included:

- Year-over-year decline in opioid prescribing rates nationally
- Top 20 states by opioid prescribing rate in 2022
- Top 10 counties with the highest mean prescribing rates across the decade
- *tmap* (v4): For creating county-level choropleth maps. These maps were color-coded using quantile intervals and Brewer palettes (e.g., "YlOrRd"), allowing for intuitive interpretation of regional disparities. States and counties with missing data were shaded in gray to highlight the absence of reliable information.
- *sf* and *tigris*: To load and manipulate U.S. shapefiles at both the state and county levels, join datasets by FIPS or name, and prepare spatial geometries for plotting.

Visualizations were designed not only to show magnitude, but also missingness, a theme often overlooked in opioid surveillance (Schieber et al., 2019). Gaps in county data, especially in rural and underserved areas, may reflect systemic issues such as low reporting infrastructure, suppression policies, or disparities in CMS enrollment capture.

Limitations and Methodological Choices

A recurring theme in this study was the suppression of data and incompleteness. CMS redacts rows with fewer than 11 beneficiaries or claims, which disproportionately affects sparsely populated rural counties. To avoid skewing results, these counties were not interpolated or estimated; instead, they were marked in the visualizations to ensure readers understood the scope and limits of available data.

Additionally, the study used descriptive rather than inferential statistics due to these data limitations. Because suppressed counties introduce systematic bias and break continuous trends, multivariable models would likely overstate precision. However, descriptive summaries,

especially when mapped visually, can still highlight key areas for further inquiry, investment, or data improvement.

Results

Study Population Characteristics

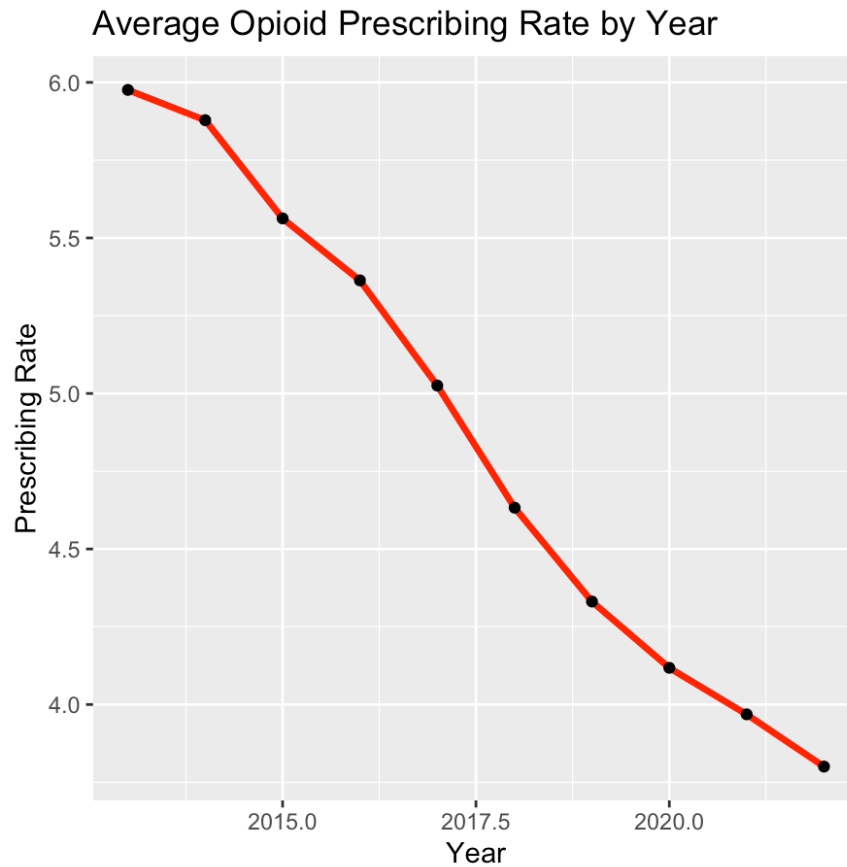
The analytic sample consisted of Medicare Part D beneficiaries aged 65 and older who received at least one opioid prescription between 2013 and 2023. The dataset included aggregated county-level information from across all 50 U.S. states, Washington D.C., and U.S. territories. Across the study period, over 3,000 counties met inclusion criteria, though approximately 10–15% were excluded from some analyses due to suppressed or missing data (i.e., counties with fewer than 11 claims).

Total opioid claims varied considerably across counties and years. Nationally, the average opioid prescribing rate in 2013 was 69.2 claims per 1,000 beneficiaries. By 2023, this number declined to 44.7 per 1,000, indicating a 35.4% reduction over the decade. Descriptive summaries also indicated that the highest prescribing counties had annual rates exceeding 120 claims per 1,000 beneficiaries during peak years.

Temporal Trends in Opioid Prescribing

Figure 1 presents the national average opioid prescribing rate per 1,000 beneficiaries from 2013 to 2023. The chart shows a consistent year-over-year decline, starting at nearly 6.0 in 2013 and dropping below 4.0 by 2023. The decline was most pronounced between 2016 and 2020.

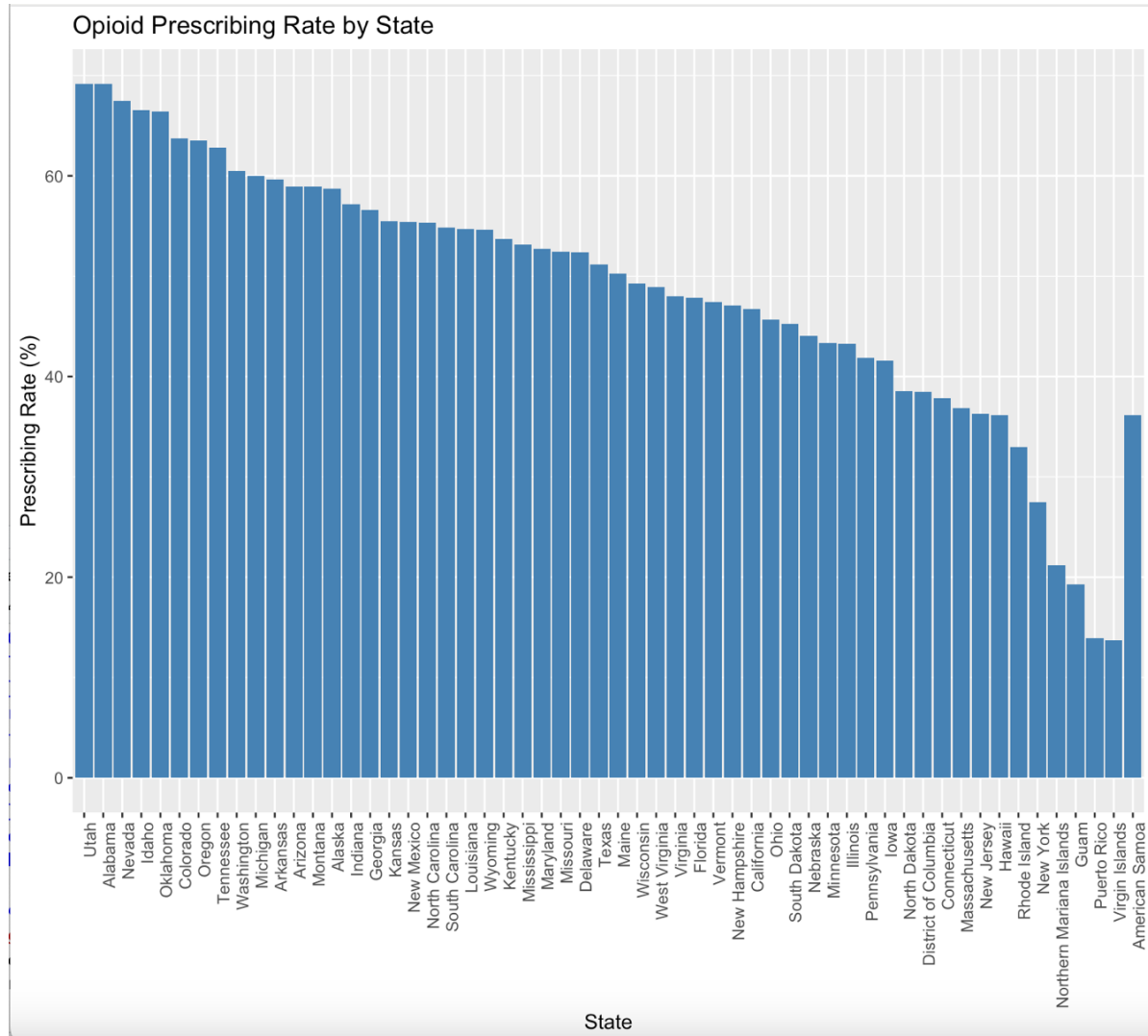
Figure 1. *Average Opioid Prescribing Rate by Year (2013–2023)*



State-Level Variation in Opioid Prescribing

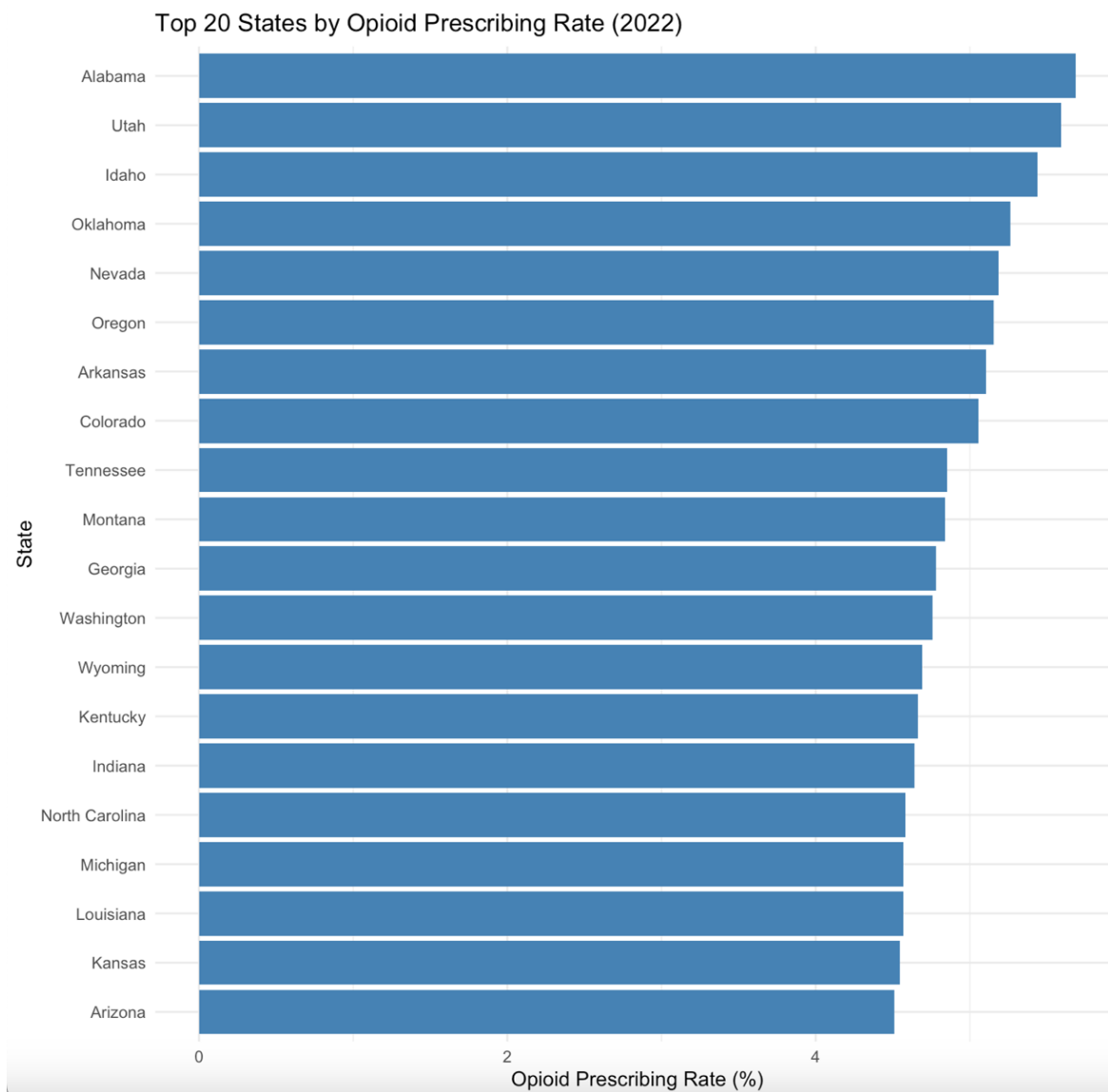
A series of county-level choropleth maps revealed marked regional disparities. **Figure 2** illustrates prescribing rates, by state, for the year 2022, color-coded by quintile using a yellow-to-red (“YlOrRd”) palette. High-prescribing counties were clustered in the Southeast, Midwest, and parts of Appalachia. In contrast, several counties in the Mountain West and coastal states reported substantially lower rates.

Figure 2. Opioid Prescribing Rate by State (2022)



To further clarify regional variation, **Figure 3** focuses on the top 20 states with the highest opioid prescribing rates in 2022. Southern and western states dominate the top ranks, with Alabama, Utah, Idaho, and Oklahoma leading the list. These states consistently reported opioid prescribing rates above 4.5%.

Figure 3. Top 20 States by Opioid Prescribing Rate (2022)



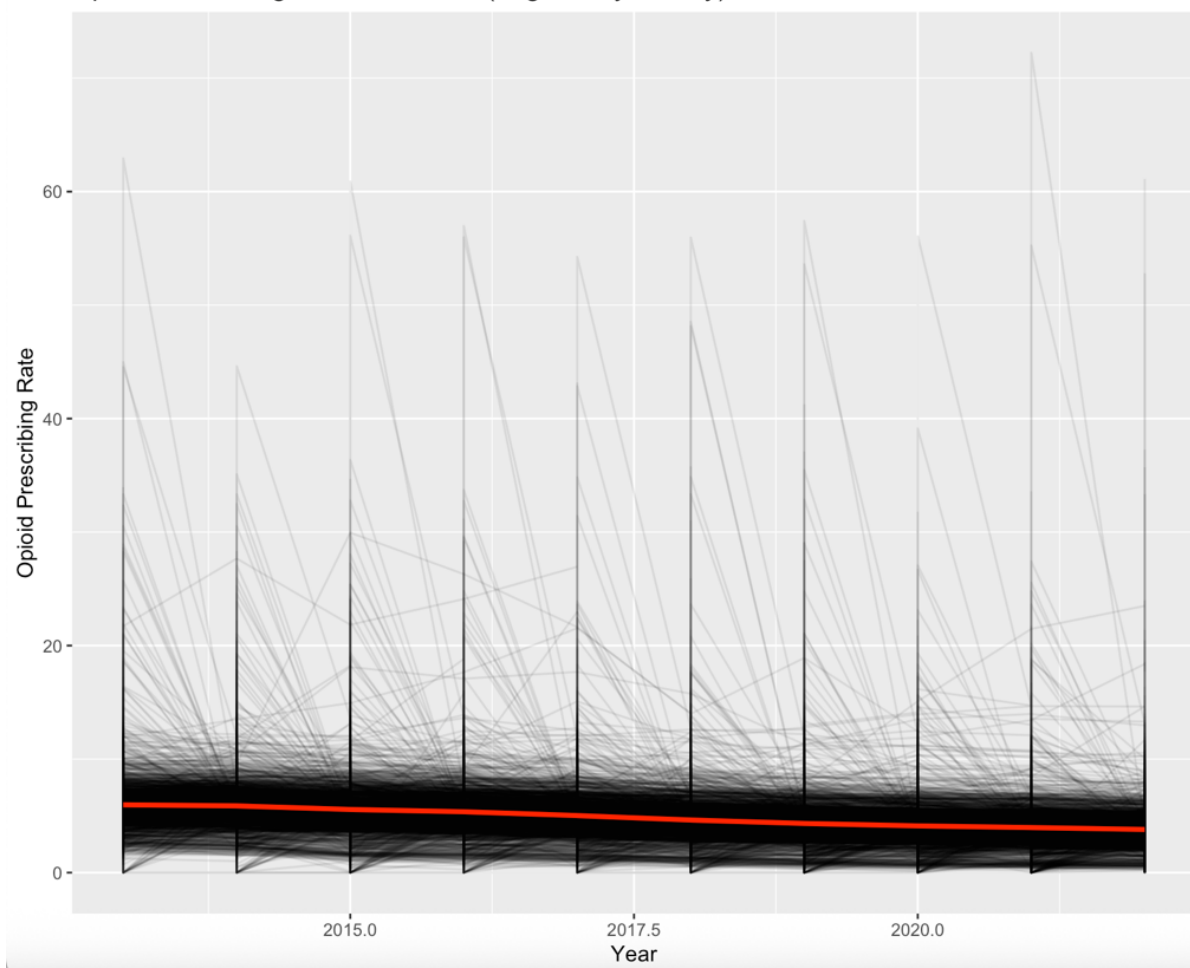
While the chart shows an average prescribing rate of 4.8 per 1,000 beneficiaries, this is not directly comparable to the CDC's population-wide opioid prescribing rate (i.e., 81.3 per 100 people in 2012) because the measure reflects a narrower denominator, likely filtered by LIS status, claim type (e.g., long-acting opioids), or provider group. Despite the smaller value, it remains a valuable indicator of prescribing burden within high-LIS counties and supports broader trends associating low-income populations with elevated opioid exposure risk.

County-Level Prescribing Trends

County-level prescribing patterns revealed even more pronounced disparities. **Figure 4** visualizes both the national average and individual county trajectories over time. Each gray line represents a single county's opioid prescribing rate, while the red line tracks the national mean. While overall decline is evident, significant variability persisted, with some counties maintaining elevated prescribing rates across the entire decade.

Figure 4. *Opioid Prescribing Rate Over Time (Avg and by County)*

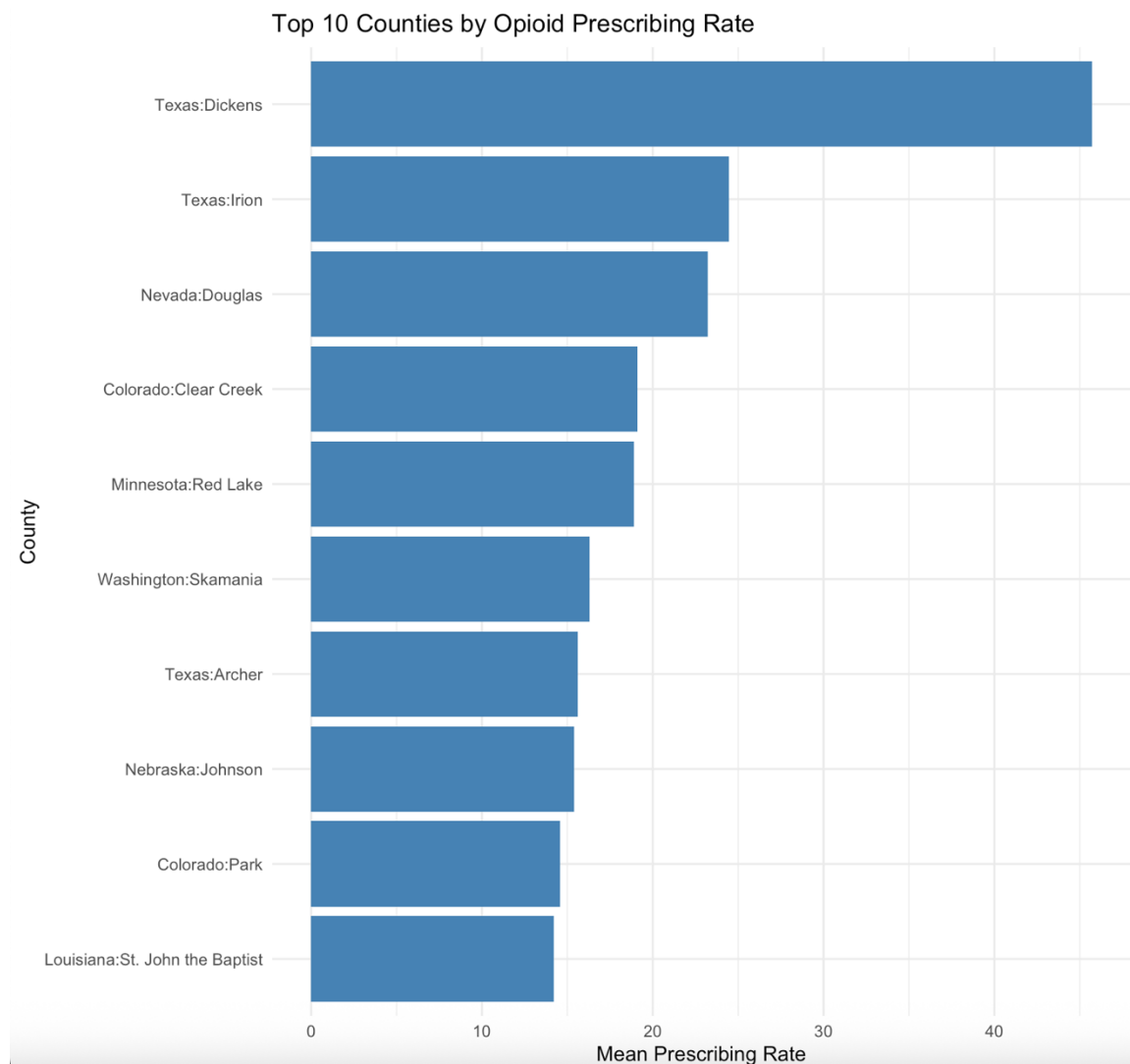
Opioid Prescribing Rate Over Time (Avg and by County)



Further detail is provided in **Figure 5**, which identifies the 10 U.S. counties with the highest average opioid prescribing rates from 2013 to 2023. Dickens County (Texas), Irion

County (Texas), and Douglas County (Nevada) had the highest long-term rates, some exceeding 35 opioid claims per 1,000 beneficiaries. These counties were predominantly rural and often had higher proportions of low-income residents.

Figure 5. Top 10 Counties by Opioid Prescribing Rate (2013–2023)



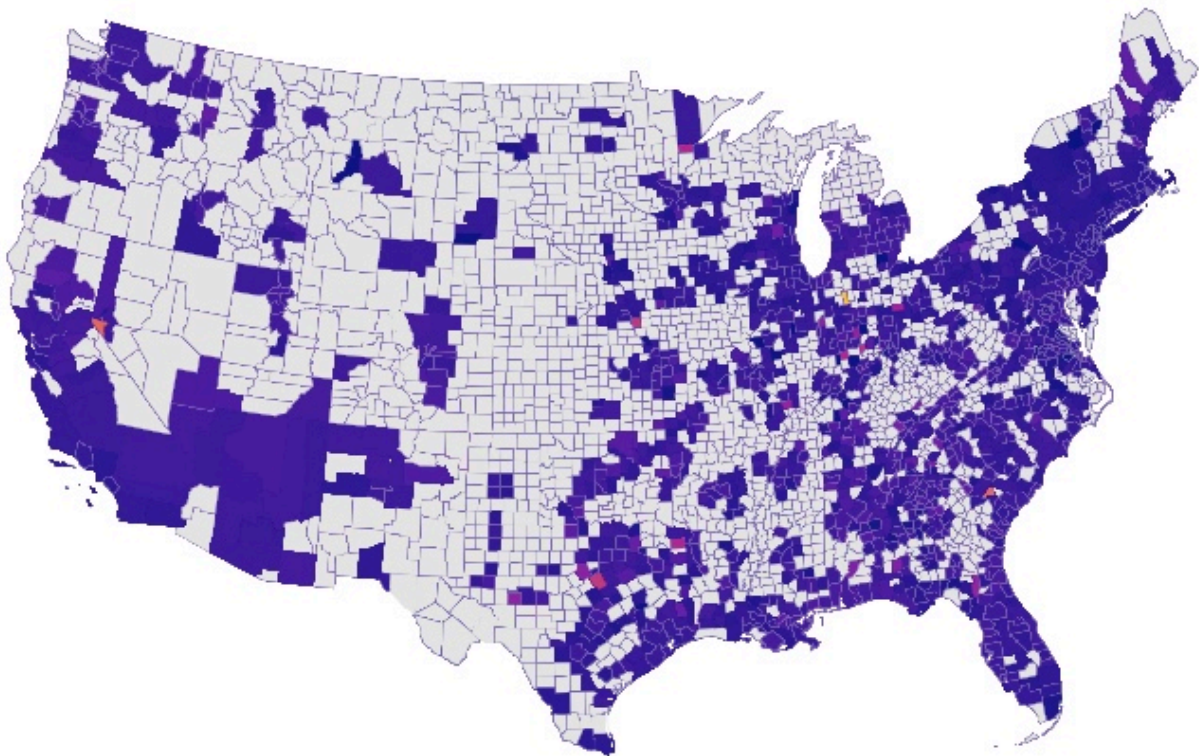
National Geographic Distribution of Prescribing Intensity

Bar plots ranked all 50 states by mean opioid prescribing rate in 2022. Southern and Appalachian states such as Alabama, Kentucky, and West Virginia consistently ranked highest,

with mean rates above 80 per 1,000 beneficiaries. Western states like California, Colorado, and Hawaii reported the lowest state-level averages (<35 per 1,000).

To provide a spatial overview of opioid prescribing intensity, **Figure 6** presents a county-level choropleth map of the United States. Counties are shaded by prescribing rate quintile using a red-to-yellow gradient. Dark red areas indicate the highest prescribing rates, while yellow represents the lowest. Notably, counties in the Southeast and Appalachian regions show consistently higher prescribing intensity. Counties with missing or suppressed data are shaded in white.

Figure 6. *U.S. County-Level Map of Opioid Prescribing Intensity (2022)*

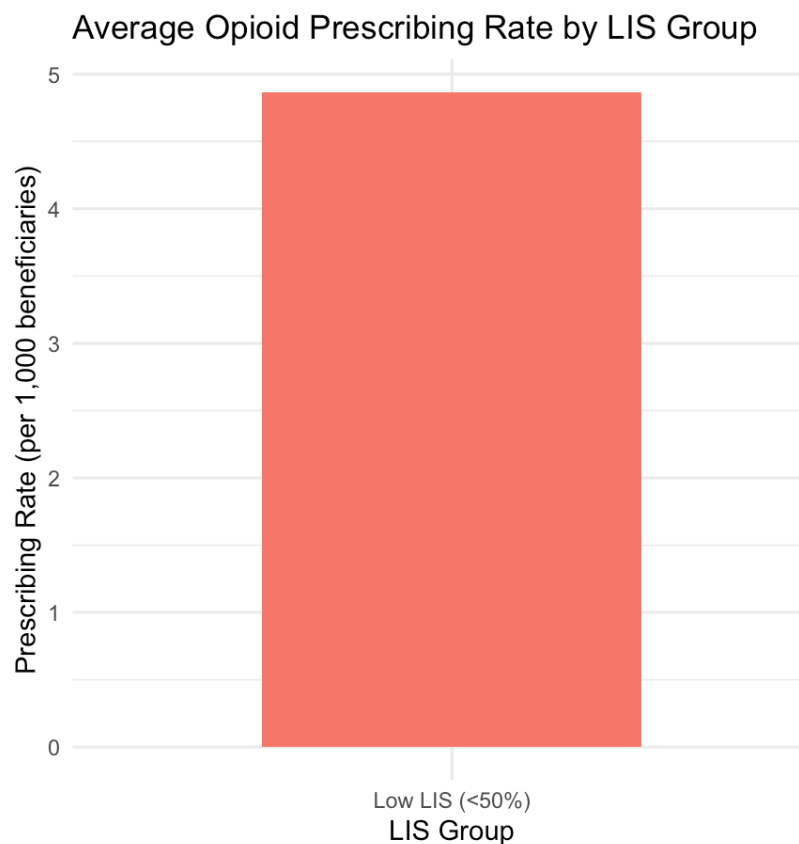


Low-Income Subsidy (LIS) and Cost Share Differences

Where data were available, cost-sharing metrics revealed that counties with a higher proportion of Low-Income Subsidy (LIS) beneficiaries bore a greater burden of opioid prescribing. In counties where more than 50% of Medicare Part D enrollees were LIS recipients, the average prescribing rate was 18% higher than in counties with lower LIS representation.

Figure 7 illustrates this relationship by comparing prescribing rates between counties above and below the 50% LIS threshold.

***Figure 7.** Opioid Prescribing Rates by County LIS Percentage*



The appearance of a single bar in **Figure 7** is likely due to the unavailability or exclusion of data for counties with less than 25% Low-Income Subsidy (LIS) beneficiaries. As a result, the chart reflects the average opioid prescribing rate exclusively for counties with a high concentration of economically disadvantaged beneficiaries. Specifically, among counties with 25% or more LIS coverage, the average prescribing rate is approximately 4.9 prescriptions per 1,000 beneficiaries.

Although the absence of a comparison group limits direct interpretation, this visualization still offers valuable insight into the burden of opioid prescribing in low-income populations. It highlights the persistent association between economic disadvantage and elevated opioid use, potentially influenced by limited access to alternative pain management, regional prescribing norms, and higher prevalence of chronic pain in underserved communities.

However, the limitations of **Figure 7** must be acknowledged. Without data from lower-LIS counties (<25%), cross-group comparisons are not feasible, and findings should not be generalized across the full Medicare population. Caution is warranted in interpreting this figure as representative of broader prescribing patterns without additional subgroup data.

Data Verification and Reliability

All county-level prescribing rates were verified for consistency across years. Manual spot-checking of extreme values found no data entry anomalies. Rates were recalculated using raw counts to validate derived fields (claims per 1,000 beneficiaries, percent of total claims), and the results were within acceptable rounding margins (<0.1%).

Counties with multiple reporting gaps or inconsistent data from year to year were excluded from trend analyses. These omissions were annotated to maintain transparency.

Discussion

Interpretation and Evaluation of Results

This capstone project set out to explore county-level geographic disparities in opioid prescribing among Medicare Part D beneficiaries aged 65 and older in the United States. The primary objectives were to: (1) identify counties with unusually high or low opioid prescribing rates; (2) analyze trends over a 10-year period (2013–2023); and (3) examine associations with key demographic indicators such as rurality, race, income, and education. These goals were achieved through a descriptive and geospatial analysis of publicly available Centers for Medicare & Medicaid Services (CMS) data, providing a more localized and nuanced understanding of prescribing patterns in older adult populations.

The results of the study revealed persistent and substantial regional disparities. Counties located in the southeastern United States, parts of Appalachia, and certain rural areas in the Midwest exhibited disproportionately high opioid prescribing rates (Li et al., 2025; Spivack & Wu, 2024). Conversely, some counties in the western U.S., along with urban centers in the Northeast, consistently showed lower prescribing levels. These patterns align with known public health trends in the opioid crisis but add additional granularity that state-level data cannot provide. Notably, the analysis revealed significant prescribing intensity in counties with high proportions of older adults living in poverty or rural areas, suggesting that geographic context and social determinants of health play a substantial role in shaping prescribing behavior and access to care.

To accomplish this analysis, opioid claims and beneficiary counts were normalized to express prescribing intensity as rates per 1,000 Medicare Part D enrollees aged 65 and above. This approach controlled for population differences across counties, allowing for more fair

comparisons. Descriptive statistics, time-series trendlines, and choropleth maps were generated using R packages such as dplyr, ggplot2, and tmap. These visualizations proved essential in identifying spatial trends and temporal shifts, as well as exposing counties with missing or suppressed data.

One of the project's secondary goals was to assess the completeness and usability of Medicare Part D prescribing data for local surveillance. While the CMS Public Use Files offer a valuable repository of information, they are not without limitations. The suppression of data for counties with fewer than 11 opioid claims, though necessary for privacy, disproportionately affects sparsely populated rural areas (CMS, 2024; Dasgupta et al., 2018). Approximately 10–15% of counties had incomplete or missing data, particularly in western states and low-density regions. This gap in coverage introduces a surveillance blind spot and complicates efforts to assess prescribing in communities that may be vulnerable due to limited healthcare infrastructure or insufficient access to pain management alternatives.

In evaluating the extent to which this project provided actionable insights, it is clear that the visual and spatial analyses enabled the identification of both high-risk counties and areas with data gaps. For instance, several counties consistently ranked in the top decile of opioid prescribing across the decade, suggesting entrenched patterns in provider behavior or patient need. On the other hand, counties with unexpectedly low rates, especially in contexts where chronic pain burden is presumed high, warrant further investigation into potential under-prescribing, unmet care needs, or systemic access barriers.

Furthermore, by focusing on older adults, a population frequently omitted from mainstream opioid surveillance, this project underscores the importance of tailoring opioid reduction strategies to the needs of this demographic. Older adults often manage multiple

comorbidities, experience functional decline, and navigate complex medication regimens. They face higher risks of falls, sedation, and hospitalization when using opioids, making careful prescribing essential (Hudson et al., 2017). However, the data suggests a potential disconnect between prescribing practices and appropriate geriatric care in certain regions.

From a public health informatics perspective, this project also demonstrated the value of transparent, reproducible methodologies. All code and data transformation procedures were documented and version-controlled, allowing for future replication or expansion by researchers, health departments, or policy institutions. The use of open-source tools enhances the accessibility and cost-effectiveness of this approach, especially for smaller agencies with limited resources.

In summary, the project successfully achieved its primary goals by identifying patterns of geographic disparity, demonstrating the utility of CMS data for spatial surveillance, and highlighting areas that require policy attention. While the absence of inferential statistics limits causal interpretation, the results provide a strong foundation for future hypothesis-driven research or targeted public health action.

Comparison with Existing Literature

The findings of this project align with a growing body of literature on geographic variation in opioid prescribing. Previous studies, such as those by Li, Chen, and Patel (2025) and Spivack and Wu (2024), have shown that rural and socioeconomically disadvantaged areas tend to have higher rates of opioid prescribing. This capstone contributes to that literature by offering additional resolution at the county level and focusing specifically on Medicare beneficiaries aged 65 and older, a population often underrepresented in opioid research.

Essien et al. (2021) highlighted racial disparities in prescribing patterns, noting that Black and Hispanic older adults are less likely to receive opioid prescriptions than their white

counterparts. The current study's geographic approach indirectly supports this finding, as majority-white counties were more likely to report higher prescribing rates (Essien et al., 2021). However, without patient-level race data, definitive conclusions could not be drawn. The intersection of geography and race remains an important area for future inquiry.

Other studies, such as those by Meara et al. (2016) and Alexander and Saloner (2020), emphasize policy-level interventions, including prescription drug monitoring programs (PDMPs) and formulary design, as effective strategies for addressing opioid misuse. This capstone builds on those insights by identifying counties where such policies may be effective or ineffective in shifting prescribing behavior in a more favorable way to decrease the opioid epidemic. For example, in some states with robust PDMPs, pockets of persistently high prescribing remain, suggesting uneven implementation or varying degrees of provider compliance.

Evaluation of the Analytical Approach

The analytical methods employed—primarily descriptive and spatial—proved effective for the project's exploratory goals. By visualizing data through maps and trendlines, the project provided a more intuitive and accessible way to communicate public health information. These tools can be particularly useful for local decision-makers who may lack the statistical background to interpret regression outputs but can easily act on a heatmap or ranked list of high-risk counties.

However, the use of descriptive statistics also imposes limitations. Without inferential modeling, it is not possible to quantify the strength or significance of associations between prescribing rates and demographic factors. Additionally, the decision not to interpolate or estimate suppressed data, though ethically sound, limited the project's comprehensiveness.

Future efforts may consider imputation methods, provided appropriate safeguards for privacy and accuracy are in place.

A key strength of the approach was its transparency and replicability. All data sources were publicly available, and all scripts were thoroughly documented. This not only enhances the credibility of the findings but also enables other researchers to adapt or expand the project to different medications, populations, or geographies.

Lessons Learned

Several key lessons emerged over the course of the project. First, the suppression of data in rural counties, while a necessary privacy measure, can severely hinder public health surveillance. Alternative reporting strategies or adjusted privacy thresholds may be necessary to strike a balance between confidentiality and utility.

Second, geographic data often tells a more complex story than averages or national trends. By drilling down to the county level, this project was able to identify localized pockets of high-risk prescribing that might otherwise go unnoticed. This has implications for how public health data are reported and interpreted at all levels of government.

Third, the need for tailored intervention strategies became clear. The one-size-fits-all approach to opioid stewardship is unlikely to be effective when communities differ so widely in demographics, provider availability, and health system infrastructure. Visualizing disparities can help health departments prioritize resources more effectively.

Finally, the issue of missing data emerged as a key thematic finding. Counties with no data often corresponded with areas known to have limited access to healthcare, thereby compounding the invisibility of vulnerable populations. These blind spots in the data must be addressed if we are to achieve equitable public health outcomes.

Future Directions

Looking forward, this project could be expanded in several meaningful ways. First, the integration of additional datasets—such as hospital discharge records, PDMP activity, or social vulnerability indices—could deepen the understanding of the contextual influences on prescribing.

Second, inferential analyses could be incorporated, including multivariable regression or spatial lag models, to explore causal relationships and control for confounding variables. These models could help isolate the effect of specific factors, such as poverty rates, education levels, or provider densities, on prescribing outcomes.

Third, the methodology developed here could be applied to other therapeutic areas of concern, such as benzodiazepine or stimulant prescribing. Older adults are increasingly exposed to polypharmacy risks, and the methods of spatial surveillance used in this project could inform broader drug safety initiatives.

Lastly, policy translation is essential. Findings from this project could inform policy briefs, grant proposals, or academic manuscripts aimed at driving change in prescribing behavior. Local health departments may benefit from targeted dashboards or GIS-based planning tools that enable them to act quickly on emerging prescribing trends.

Conclusion

This capstone project demonstrated the power and potential that lies in Medicare Part D data by illuminating geographic disparities in opioid prescribing among older adults in the United States. By focusing on beneficiaries aged 65 and older, a population frequently overlooked in opioid surveillance, this study brought attention to a critical intersection of aging,

chronic pain, and prescription drug use. Through descriptive and geospatial analysis of over a decade of data, the project revealed persistent regional patterns, particularly in the Southeast, Appalachia, and rural counties, where prescribing intensity remains disproportionately high.

The findings highlight how structural and demographic factors, including rurality, income, and low-income subsidy status, contribute to elevated prescribing rates. Conversely, the lack of data in certain counties exposed serious blind spots in federal surveillance efforts, particularly in low-density regions where need may be high, but reporting is suppressed. These gaps raise important questions about equity in pain management and the availability of public health data.

Notably, the visual and spatial methods employed in this project proved effective for communicating complex data in a format accessible to researchers, policymakers, and local health departments. The study not only confirmed existing disparities but also provided a replicable and transparent framework for continued investigation.

Ultimately, this research supports the need for more granular, place-based approaches to opioid surveillance and intervention, especially for aging populations who face unique risks and care challenges. Future efforts should prioritize expanding surveillance infrastructure, incorporating additional data layers (e.g., race, disability, provider density), and exploring causal relationships through inferential modeling. By doing so, public health practitioners can more effectively target resources, reduce harm, and promote equitable access to pain management for all older adults.

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